

trial wireless control, IP cameras, sensor networks, smart power plugs, baby monitors, wearable electronics, security ID tags, position system beacons and location-aware devices.

Software

In this project we have used Arduino IDE for compiling and loading programs. ESP8266 community has developed a suitable plugin for ESP8266 to use with Arduino IDE. The plugin supports three types of ESP8266 Wi-Fi boards. These are generic ESP8266 board, node MCU ESP8266 board and OLIMEX ESP8266 development board.

Installing the ESP8266 Arduino

Addon. Install Arduino IDE (1.6.5 version). Launch Arduino IDE and open preferences window from File→Preferences option (Fig. 5). Choose Additional Boards Managers URLs: field and enter http://arduino.esp8266.com/stable/package_esp8266com_index.json. Click OK.

Open Boards Manager from Tools→Board: Arduino Uno→Boards Manager and scroll down to find ESP8266 by ESP8266 Community. Highlight the field and Install button will appear. Click it. The process will take nearly 30 minutes with an Internet connection.

The software is simple. `setup()` routine initialises the hardware by configuring the serial port, GPIO2 pin as output and GPIO12 as input. `setupWiFi()` routine configures the wireless section. `loop()` routine handles requests from the client.

Function `size_t sendProgmem(WiFiClient client, const char progmem[], size_t size)`, sends data to the client. array `IN[ ]` corresponds to the Web page (Fig. 6), `image[ ]` array corresponds to EFYLOGO (Fig. 7) and `ES[ ]` array corresponds to the module (Fig. 2).

Arrays are created using file-to-array converter software hexy. It is free and can be downloaded from the Internet. While creating the array, remove image extension jpg for

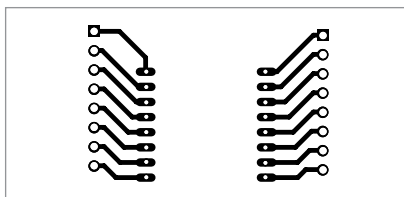


Fig. 10: Actual-size PCB pattern of the child board

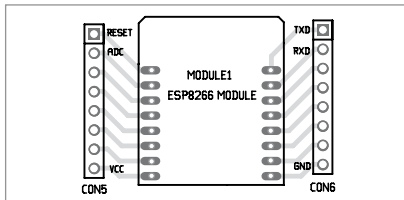


Fig. 11: Component layout of the child board



Fig. 12: USB-to-serial UART converter

decoding purpose. All jpg files have been included in this month's DVD.

Suitable routines are included to change and store SSID and password. After configuration of ESP8266 is done for Wi-Fi, an 8-bit serial shift to parallel-shift register program using HCF4094 is used for LCD interface.

Construction and testing

An actual-size, single-side PCB for the main board is shown in Fig. 8 and its component layout in Fig. 9. An actual-size, single-side PCB for the child board is shown in Fig. 10 and its component layout in Fig. 11. IC6 to be soldered to the bottom side after proper identification.

Install USB-to-serial UART converter (Fig.12) driver software and perform loopback test.

Select Generic ESP8266 Board

Module from Tools→Board: Arduino Uno. Arduino IDE will accept your board and you can fuse the program using Upload button. To fuse the program into the module, connect the COM port (CON2) of the main board to your PC through serial-to-USB converter and select as well as note down the emulating virtual COM port number.

Power-on the main board. Holding Program button S1 (GPIO0) down, press Reset button (S2) down. Release Reset and then release S1 (GPIO0) button. Now ESP8266 will enter into program mode. Click Upload button and the program will get compiled and generated bin file will load into the module.

Power-off the main board and power-on your Wi-Fi dongle. Open hyperterminal (X-CTU hyperterminal has been used in EFY Lab) on the PC with baud rate 115200 and select COM port. Hold S3 down and power-on the main board. You will get a prompt for entering SSID and password. Release switch and enter SSID and password of your Wi-Fi network separated by comma. Click Enter. SSID and password will be stored in the EEPROM.

For subsequent use, you need not use S3. When Wi-Fi is connected, IP address of ESP8266 module will be displayed in LCD1. Launch the browser with the IP address. You will get a Web page as shown in Fig. 6. Buttons marked as LEDON and LEDOFF are for switching on and off LED1. Appliances like fans and motors can be switched on and off with a suitable relay driver circuit using optocouplers connected to pin 11 (GPIO2). Status of pin 6 (GPIO12) and temperature readings are displayed using AJAX scripts without refreshing the entire Web page. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

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